

Micron Technology, Inc.

Recommendation BUY

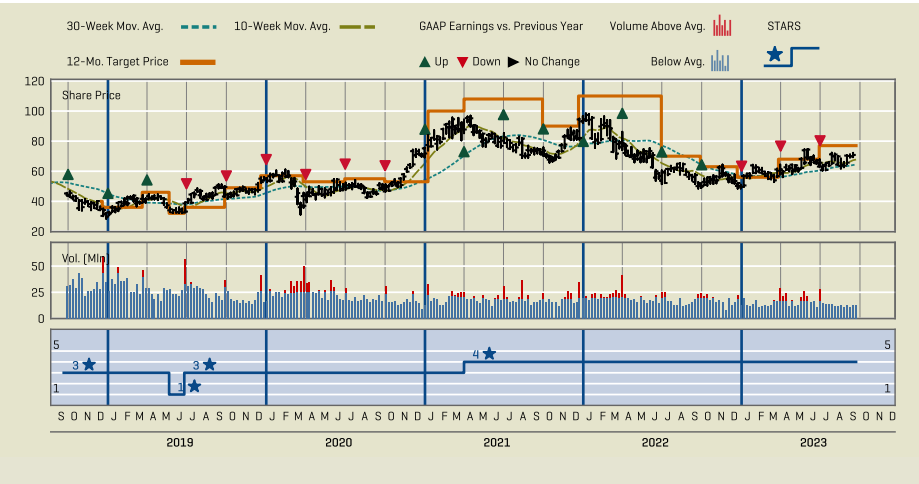
Price USD 68.58 [as of market close Sep 25, 2023] 12-Mo. Target Price USD 77.00 Report Currency USD Investment Style Large-Cap Value

Equity Analyst Angelo Zino, CFA

GICS Sector Information Technology Sub-Industry Semiconductors Summary Micron Technology is a manufacturer of semiconductor memory products, including DRAM and NAND flash memory, as well as image sensors.

Key Stock Statistics [Source: CFRA, S&P Global Market Intelligence (SPGMI), Company Reports]					
52-Wk Range	USD 74.77 - 48.43	Oper.EPS2023E	USD -4.57	Market Capitalization[B]	USD 75.12
Trailing 12-Month EPS	USD -1.93	Oper.EPS2024E	USD 0.16	Yield [%]	0.67
Trailing 12-Month P/E	NM	P/E on Oper.EPS2023E	NM	Dividend Rate/Share	USD 0.46
USD 10K Invested 5 Yrs Ago	15,401.0	Common Shares Outstg.[M]	1,095.00	Trailing 12-Month Dividend	USD 0.34
				Beta	1.36
				3-yr Proj. EPS CAGR[%]	-21
				SPGMI's Quality Ranking	B
				Institutional Ownership [%]	83.0

Price Performance



Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by Angelo Zino, CFA on Sep 26, 2023 10:01 AM, when the stock traded at USD 67.07.

Highlights

- We think sales will rebound 37% in FY 24 [Aug.] and 41% in FY 25 after our view of a 50% decline in FY 23. We see sharply higher demand in CY 24, largely supported by mass production of its high-bandwidth memory (HBM3) offering as well as greater penetration for its DDR5 chips [more than doubled in May-Q from Feb-Q; DDR5 seen crossing over DDR4 in Q1 of CY 24]. We expect pricing to trend higher over the next six to eight quarters and think higher orders from key smartphone/PC vendors will improve visibility and support an industry recovery.
- We forecast an annual gross margin of 18% in FY 24 and 33% in FY 25 versus a 4% projection in FY 23. We expect pricing to stabilize/improve after sharp declines in recent quarters, with fab utilization improving as demand returns. Node transitions will aid costs over time, while write-downs abate.
- We note \$11.3B of cash and \$13.2B debt, with MU to hold off on buybacks amid macro concerns. We view lower fab utilization/capex cuts as prudent [-40% in FY 23, WFE spend -50%] and like output cuts from peers. We think wafer start reductions [-30%] are warranted as MU focuses on managing down inventories [DIOs targeted at 120 vs. 168 current] and controlling supply.

Investment Rationale/Risk

- Our Buy reflects our view of valuation and belief that the memory industry is poised to recover. While China exposure is a risk via a failed cybersecurity review [about a quarter of total sales; half of that at risk], greater memory content in AI servers offer a long-term secular opportunity [DRAM has 6x-8x and NAND 3x the memory of traditional workloads]. We like the emphasis on Compute Express Link [CXL] memory products, and see higher volume, cost-per-bit reductions, and an improving mix in the long term. Secular prospects in data centers remain bright, led by higher cloud spend, while new server CPUs will feature more memory channels that will aid server memory demand. Higher priced DDR5 offerings are gaining traction, while HBM3 poses upside potential to our forward assumptions.
- Risks: lower-than-expected sales of computers and consumer electronic products, faster-than-expected capacity expansion in the memory industry, and less-than-anticipated cost cuts.
- Our 12-month target of \$77 is based on our P/E analysis. We apply a P/E of 13x to our FY 25 EPS estimate of \$5.91, which is below peers, but near the five-year forward historical average of 11.7x to reflect our view of earnings volatility.

Update: please see the analyst's latest research note in the research notes section

Analyst's Risk Assessment

LOW	MEDIUM	HIGH
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Micron is subject to semiconductor industry cyclicality and sudden changes in pricing for commodity memory products. It is a relatively large semiconductor company and is the lone American survivor in the global DRAM industry, which has been consolidating in recent years.

Revenue/Earnings Data

Revenue (Million USD)					
	1Q	2Q	3Q	4Q	Year
2025	E 7,014	E 7,150	E 7,500	E 8,222	E 29,888
2024	E 4,437	E 4,865	E 5,548	E 6,325	E 21,175
2023	4,085	3,693	3,752	E 3,900	E 15,430
2022	7,687	7,786	8,642	6,643	30,758
2021	5,773	6,236	7,422	8,274	27,705
2020	5,144	4,797	5,438	6,056	21,435

Earnings Per Share (USD)

	1Q	2Q	3Q	4Q	Year
2025	E 1.17	E 1.35	E 1.69	E 1.70	E 5.91
2024	E -0.78	E -0.20	E 0.30	E 0.84	E 0.16
2023	-0.04	-1.91	-1.43	E -1.19	E -4.57
2022	2.16	2.14	2.59	1.45	8.35
2021	0.78	0.98	1.88	2.42	6.06
2020	0.48	0.45	0.82	1.08	2.83

Fiscal Year ended Aug 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data

Amount [USD]	Date Decl.	Ex-Div. Date	Stk. of Record	Payment Date
0.1150	Jun 28	Jul 07	Jul 10	Jul 25 '23
0.1150	Mar 28	Apr 06	Apr 10	Apr 25 '23
0.1150	Dec 19	Dec 30	Jan 03	Jan 19 '23
0.1150	Sep 29	Oct 07	Oct 11	Oct 26 '22

Dividends have been paid since 1991. Source: Company reports
Past performance is not an indication of future performance and should not be relied as such.
Forecasts are not a reliable indicator of future performance.
Dividends paid in currencies other than the Trading currency have been accordingly converted for display purposes.

Micron Technology, Inc.**Business Summary** June 29, 2023

CORPORATE OVERVIEW. Micron Technology is a manufacturer and marketer of dynamic random access memory (DRAM) and NAND flash memory. The company's products are used in an increasingly broad range of electronic devices, including personal computers, network servers, mobile phones, digital still cameras, and other consumer electronic products. Micron is one of the largest memory makers, ranking among the top five suppliers in the world.

DRAM products are dynamic random access memory semiconductor devices with low latency that provide high-speed data retrieval with various performance characteristics. DRAM products lose content when power is turned off (volatile) and are most commonly used in client, cloud server, enterprise, networking, graphics, industrial, and automotive markets. Low-power DRAM (LPDRAM) products, which are engineered to meet standards for performance and power consumption, are sold into smartphone and other mobile-device markets, as well as into the automotive, industrial, and consumer markets.

NAND products are non-volatile, re-writeable semiconductor storage devices that provide high-capacity, low-cost storage with various performance characteristics. NAND is used in SSDs for the enterprise and cloud, client, consumer markets, and removable storage markets. Managed NAND is used in smartphones and other mobile devices, as well as in consumer, automotive, and embedded markets. Low-density NAND is used for applications such as automotive, surveillance, machine-to-machine, automation, printer, and home networking.

Micron operates under four business units: 1) Compute and Networking, which includes memory sold into compute, networking, graphics, and cloud server markets; 2) Storage, which includes memory and storage sold into enterprise, client, cloud, and removable storage markets; 3) Mobile, which includes memory products sold into smartphone, tablet, and other mobile-device markets; and 4) Embedded, which includes memory sold into the automotive, industrial, connected home, and consumer electronics markets.

MARKET PROFILE. We believe CY 2022 industry bit demand grew in the low- to mid-single-digit percentage range for both DRAM and NAND. CY 2022 industry bit supply growth for both DRAM and NAND is seen being close to their respective long-term demand CAGRs and well above CY 2022 industry demand growth. An unprecedented confluence of macro events and customer inventory adjustments are depressing demand for DRAM and NAND to well below end consumption. We also note an extremely aggressive pricing environment near term, as we expect supply growth to be significantly above the demand growth, contributing to very high supplier inventories for both DRAM and NAND.

For CY 2023, Micron expects industry demand growth of about 4%-6% in DRAM and about 9% in NAND. For CY 2023, Micron projects that DRAM and NAND industry supply growth will be well below their long-term CAGRs and also well below expected demand growth in 2023. Given the significant supply demand mismatch entering CY 2023, industry profitability will remain challenged throughout 2023.

Longer term, Micron sees DRAM bit demand CAGR in the mid-teens percentage range and NAND bit demand CAGR in the low-20% range. The memory market has been on a virtual roller coaster of highs and lows, affected by volatile end market demand, inventory levels, price wars, and advances in technology and manufacturing. Sales tend to have very large swings, and because of the high-fixed cost nature of the memory industry, earnings are even more erratic.

Commodity memory chips represent one of the largest and most volatile segments of the semiconductor industry. Memory sales have risen annually, as higher bit shipments have offset deteriorating prices. Historically, DRAM prices have declined more than 20% annually, on average. However, DRAM prices have a very wide range and prices have plunged as much as 75%, or increased as much as 200% in a matter of months. NAND prices have also experienced wild swings over the years but trends lower long term.

The company faces competition in the semiconductor memory and storage markets from various companies, including Kioxia Holdings Corporation (formerly Toshiba Memory Corp.), Samsung Electronics, SK Hynix, and Western Digital.

CORPORATE STRATEGY. Manufacturing technology is a key element in the company's corporate strategy, we believe, because it allows it to make smaller and more dense memory chips that are attractive to customers looking to reduce the size of their products. Further, manufacturing advances lead to cost reductions on a per bit basis, supporting profitability in the face of pricing deterioration.

Semiconductor manufacturing is highly capital intensive, requiring large investments in new facilities and equipment. Most of Micron's semiconductor equipment must be replaced every three to five years with increasingly advanced equipment. Therefore, it is important for the company to gain or maintain market share and increase its exposure to faster-growing segments within the memory markets, we think, in order to expand sales, improve margins, and maintain a healthy return on investment.

To share the costs of building and operating expensive manufacturing plants, the company has entered into a number of joint ventures with various companies.

FINANCIAL TRENDS. Micron's sales are not only cyclical but are highly sensitive to industry supply and demand conditions. Revenues increased 11% in FY 2022 after rising 29% in FY 2021 and after falling 8% in FY 2020. For FY 2022, DRAM revenue increased 12% to \$22.4 billion, representing 73% of total fiscal year revenue. NAND revenue increased 11% year-over-year to a record \$7.8 billion. We note that because of the sales variability each year, margins have fluctuated widely. Gross margins have been very volatile, ranging from 59% to -9% since 2006. Even more volatile, operating margins were anywhere from 49% to -35% over this time frame.

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Executive Vice President of**Technology & Products**

S. J. DeBoer

VP, Interim General**Counsel & Corporate****Secretary**

M. Myers

Independent Chairman of the Board

R. E. Switz

Corporate VP & Chief**Accounting Officer**

S. R. Allen

CEO, President & Director

S. Mehrotra

Board Members

L. A. Dugle

R. E. Switz

L. M. Haynesworth

R. M. Beyer

M. A. Wright

S. J. Gomo

M. P. McCarthy

S. Mehrotra

Domicile

Delaware

Auditor

PricewaterhouseCoopers LLP

Founded

1978

Employees

48,000

Stockholders

1,768

Micron Technology, Inc.

Quantitative Evaluations						Expanded Ratio Analysis					
Fair Value Rank	NR	1	2	3	4	5	2022	2021	2020	2019	
LOWEST						HIGHEST	Price/Sales	2.09	3.05	2.44	2.18
Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued [1] to most undervalued [5].							Price/EBITDA	3.82	6.49	6.02	3.98
							Price/Pretax Income	6.72	13.50	17.52	7.24
							P/E Ratio	6.86	12.21	16.37	7.03
							Avg. Diluted Shares Outstg. [M]	1,122.00	1,141.00	1,131.00	1,143.00
						Figures based on fiscal year-end price					
Fair Value Calculation	N/A										
Volatility	LOW		AVERAGE		HIGH						
Technical Evaluation	BULLISH	Since September, 2023, the technical indicators for MU have been BULLISH"									
Insider Activity	UNFAVORABLE		NEUTRAL		FAVORABLE						
						Key Growth Rates and Averages					
						Past Growth Rate [%]		1 Year	3 Years	5 Years	
						Net Income		48.22	11.23	11.29	
						Sales		11.02	9.53	8.64	
						Ratio Analysis [Annual Avg.]					
						Net Margin [%]		28.24	20.64	27.08	
						% LT Debt to Capitalization		10.47	11.67	10.67	
						Return on Equity [%]		18.51	13.26	22.31	

Company Financials Fiscal year ending Aug 31										
Per Share Data [USD]	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
Tangible Book Value	44.11	37.85	33.63	31.02	26.47	15.29	11.07	11.39	9.59	8.39
Free Cash Flow	2.80	2.18	0.07	3.06	7.40	3.14	-2.56	1.11	2.45	0.36
Earnings	7.75	5.14	2.37	5.51	11.51	4.41	-0.27	2.48	2.54	1.13
Earnings [Normalized]	8.35	6.06	2.83	6.35	11.95	4.96	0.06	2.69	3.23	-0.12
Dividends	0.43	0.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Payout Ratio [%]	5.31	NM	NM	NM	NM	NM	NM	NM	NM	NM
Prices: High	98.45	96.96	61.19	53.68	64.66	32.96	19.30	36.59	34.85	15.07
Prices: Low	51.40	44.45	31.13	28.39	31.38	16.14	9.31	13.50	13.06	5.16
P/E Ratio: High	11.80	16.00	21.60	8.50	5.40	6.60	NM	13.60	10.80	NM
P/E Ratio: Low	6.20	7.30	11.00	4.50	2.60	3.30	NM	5.00	4.00	NM
Income Statement Analysis [Million USD]										
Revenue	30,758	27,705	21,435	23,406	30,391	20,322	12,399	16,192	16,358	9,073
Operating Income	9,709	6,796	3,060	7,390	14,926	5,877	234.00	2,984	3,370	450.00
Depreciation + Amortization	7,116	6,214	5,650	5,424	4,759	3,861	2,980	2,667	2,103	1,804
Interest Expense	189.00	183.00	194.00	128.00	342.00	601.00	437.00	371.00	352.00	231.00
Pretax Income	9,575	6,255	2,990	7,051	14,306	5,204	-256.00	3,056	3,207	1,202
Effective Tax Rate	9.30	6.30	9.40	9.80	1.20	2.20	-7.40	5.10	4.00	0.70
Net Income	8,687	5,861	2,687	6,313	14,135	5,089	NM	2,899	3,045	1,190
Net Income [Normalized]	6,024	4,179	1,843	4,621	9,136	3,275	NM	1,932	2,130	85.40
Balance Sheet and Other Financial Data [Million USD]										
Cash	9,331	8,633	8,142	7,955	6,802	5,428	4,398	3,521	4,534	3,101
Current Assets	21,781	19,907	17,965	16,503	16,039	12,457	9,495	8,596	10,245	8,911
Total Assets	66,283	58,849	53,678	48,887	43,376	35,336	27,540	24,143	22,416	19,118
Current Liabilities	7,539	6,424	6,635	6,390	5,757	5,355	4,835	3,905	4,791	4,125
Long Term Debt	6,020	5,972	5,963	4,173	3,241	9,039	8,128	5,835	4,373	3,607
Total Capital	57,483	51,268	46,226	42,719	37,900	30,625	22,838	20,629	18,141	16,043
Capital Expenditures	12,067	10,030	8,223	9,780	8,879	4,734	5,817	4,021	3,107	1,442
Cash from Operations	15,181	12,468	8,306	13,189	17,400	8,153	3,168	5,208	5,699	1,811
Current Ratio	2.89	3.10	2.71	2.58	2.79	2.33	1.96	2.20	2.14	2.16
% Long Term Debt of Capitalization	10.50	11.60	12.90	9.80	8.60	29.50	35.60	28.30	24.10	22.50
% Net Income of Revenue	28.20	21.20	12.50	27.00	46.50	25.00	-2.20	17.90	18.60	13.10
% Return on Assets	9.70	7.55	3.73	10.01	23.70	11.68	0.57	8.01	10.14	1.68
% Return on Equity	18.50	14.10	7.10	18.10	53.60	31.40	-2.10	23.40	28.60	13.00

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

Micron Technology, Inc.**Sub-Industry Outlook**

We have a positive fundamental outlook for the semiconductors sub-industry over the next 12 months. We expect semiconductor sales to rebound about 10% in 2024 after our forecast for a 12% decline in 2023, with a cyclical recovery to start in the second half of the year [Y/Y revenue growth returns in Q4 2023]. We are starting to see a return to sequential growth across the more cyclical areas of the semiconductor market (PCs, smartphones, and gaming), driven in part by seasonality as well as a more normalized inventory channel after sharply undershipping end-demand for the last several quarters. Whether or not recent strength can be sustained will be dependent on macro factors and the health of the consumer, but we think the industry is in a very healthy situation.

Currently, the biggest question mark is within the industrial/automotive markets as those areas held up well through 1H 2023, but channel inventory is now being rightsized due to end-market softness, which may continue until at least Q1 2024. Although inventories are still above optimal levels, companies are making efforts (e.g., production cuts) to quickly get back to normalized levels. In the long term, we expect content growth in automobiles to sharply outpace semiconductor industry growth, driven by several megatrends, such as autonomy, electrification, and connected infotainment. Analog semiconductors are particularly exposed to these trends.

Although data center spending has been relatively muted in 2023 [we estimate -2%], we see a 10%-15% rebound in 2024, with investments continuing to be focused on upgrading the network to meet more artificial intelligence [AI] driven workloads. As a result, we expect accelerated computing technologies (GPUs, FPGAs, and ASICs) to gain traction, in our view, as they help overcome the limitations of CPUs. We believe that the recent spend shift towards accelerators will be permanent and will grow to an annual run rate of over \$130 billion by 2027 versus our estimated \$45 billion-\$50 billion run rate in 2023. Chipmakers tied to the data center are exposed to the best financially positioned

customers on the planet, and despite quarterly lumpiness, the long-term prospects related to themes like AI and the metaverse remain extremely encouraging and offer significant growth potential ahead.

We believe the semiconductor industry will witness a super cycle over the next decade as content growth per device across a host of markets accelerates, driven by the emergence of greater AI capabilities. Given this, we estimate the semiconductors industry will grow at a 10-year annualized pace of 6%-7% [using 2021 expected revenue as the base year], or at least 2x global GDP.

While companies that outsource manufacturing would typically have more stability in gross margins (a trade-off for capped upside), others with their own manufacturing see more variability. Industry margins are a function of manufacturing utilization and inventory supply-demand imbalances. We forecast some companies, especially in analog, to maintain higher inventory levels. Healthy free cash flow generation and high cash levels will likely result in greater shareholder return via share repurchases. Regulatory and geopolitical uncertainties remain notable risks.

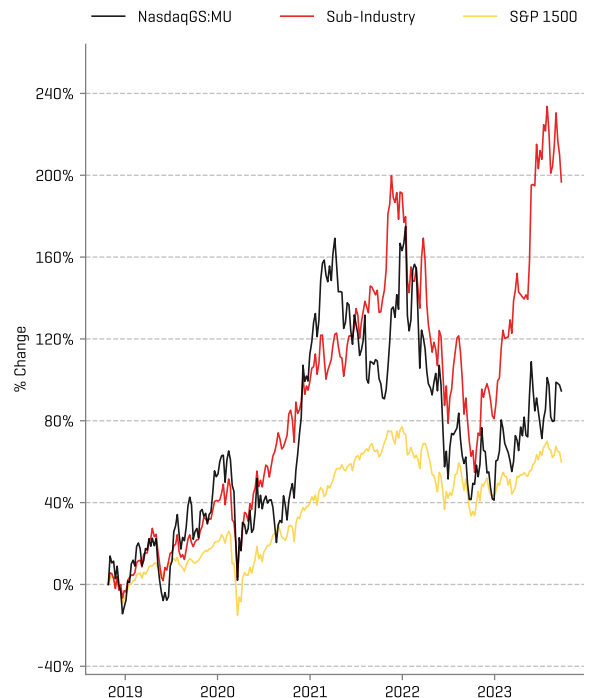
YTD through September 22, the S&P semiconductors sub-industry index rose 64.0% vs. an 11.6% increase for the S&P 1500 index. For full-year 2022, the S&P semiconductors sub-industry index fell 37.9% vs. a 19.1% decline for the S&P 1500 index.

/ Angelo Zino, CFA

Industry Performance**GICS Sector: Information Technology
Sub-Industry: Semiconductors**

Based on S&P 1500 Indexes

Five-Year market price performance through Sep 23, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.

All Sector & Sub-Industry information is based on the Global Industry Classification Standard (GICS).

Past performance is not an indication of future performance and should not be relied upon as such.

Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Semiconductors Peer Group*: Semiconductors

Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. [M]	30-Day Price Chg. [%]	1-Year Price Chg. [%]	P/E Ratio	Fair Value Calc.	Yield [%]	Return on Equity [%]	LTD to Cap [%]
Micron Technology, Inc.	MU	NasdaqGS	USD	68.88	75,444.0	5.7	38.7	NM	N/A	0.7	-6.1	19.9
Analog Devices, Inc.	ADI	NasdaqGS	USD	175.48	87,444.0	-1.1	20.7	16.0	152.68	2.0	10.3	15.0
Arm Holdings plc	ARM	NasdaqGS	USD	51.32	52,658.0	N/A	N/A	132.0	N/A	N/A	N/A	N/A
GLOBALFOUNDRIES Inc.	GFS	NasdaqGS	USD	56.83	31,408.0	4.0	3.6	21.0	N/A	N/A	15.6	16.5
Marvell Technology, Inc.	MRVL	NasdaqGS	USD	52.30	45,124.0	-15.0	20.2	31.0	N/A	0.5	-2.4	15.8
Microchip Technology Incorporated	MCHP	NasdaqGS	USD	77.08	41,957.0	-4.4	21.8	12.0	67.97	2.1	37.4	35.9
Monolithic Power Systems, Inc.	MPWR	NasdaqGS	USD	449.29	21,466.0	-13.5	17.7	36.0	406.23	0.9	27.8	N/A
NXP Semiconductors N.V.	NXPI	NasdaqGS	USD	196.79	50,733.0	-2.1	26.8	14.0	N/A	2.1	35.8	51.8
ON Semiconductor Corporation	ON	NasdaqGS	USD	93.81	40,482.0	1.0	44.4	18.0	91.45	N/A	31.5	23.6
QUALCOMM Incorporated	QCOM	NasdaqGS	USD	107.68	120,171.0	-3.1	-12.9	11.0	131.21	3.0	47.0	40.2
STMicroelectronics N.V.	STM	NYSE	USD	42.72	38,750.0	-9.7	24.6	9.0	N/A	0.6	38.4	13.9

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.

NA-Not Available; NM-Not Meaningful.

Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

Micron Technology, Inc.

Analyst Research Notes and other Company News

September 26, 2023

10:02 am ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 68.22****]

Ahead of Aug-Q results after the close on 9/27, we keep our per share estimates for FY 23 [Aug.] at a \$4.57 loss, FY 24 at \$0.16 EPS, and FY 25 at \$5.91. Although we see sales down 41% in Aug-Q, we believe MU has officially passed the cyclical trough and we see a return to Y/Y growth in Nov-Q [+3% forecast]. We believe the primary focus of the call will be on the trajectory of pricing/margins, as customer inventories are improving with signs of higher pricing being seen. Although utilization rates will remain a margin headwind through at least mid-CY 24, we expect MU to remain extremely prudent with its cost structure/capex as it looks to get back to positive FCF (we estimate May-Q). The bull case remains on the AI server side, as HBM helps support a 6x-8x content boost vs. traditional servers, but note the content growth story in consumer markets is fading [e.g., smartphones, PCs] as hardware vendor focus shifts to cost. We hold our 12-month target price at \$77 on a P/E of 13x our FY 25 EPS estimate.

/ Angelo Zino, CFA

June 29, 2023

08:42 AM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 67.07****]:

We boost our 12-month target to \$77 from \$68 on a P/E of 13x our FY 25 EPS view of \$5.91 [up from \$5.11], above historical given trough fundamentals. We widen our FY 23 [Aug.] per share loss estimate to \$4.57 from \$3.37 and cut FY 24 to \$0.16 EPS from \$1.26, hurt by China restrictions. MU posts May-Q loss per share of \$1.43 vs. \$2.59 EPS, beating the \$1.57 consensus loss view. Sales rose 2% from Feb-Q [-57% Y/Y], beating expectations, as a +14% in NAND was partly offset by a -2% in DRAM. Gross margin improved to -16% vs. -31% in Feb-Q, still hurt by \$400M in write-downs [11 pp headwind]. We see higher demand in CY 24, aided by the ramp of high-bandwidth memory as well as greater penetration for DDR5 [will surpass DDR4 in early CY 24]. Greater memory content in AI servers [DRAM has 6x-8x and NAND 3x the memory of traditional workloads] and higher pricing offer upside/better visibility. We like further wafer start reductions [-30%] with focus on managing down inventories [DIOs targeted at 120 vs. 168 current]. / Angelo Zino, CFA

May 22, 2023

07:18 AM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 64.75****]:

China says that MU has failed a network cybersecurity review and MU will be prevented from selling its memory for infrastructure purposes in the region [e.g., data centers, telecom]. We believe the move is all political and a way for China to retaliate against recent moves by the U.S. [adding Yangtze Memory to entity list; restricting NVIDIA's H100 for AI purposes]. Although China represents about 10%-11% of MU's sales [16% with Hong Kong], we expect the overall impact to be less than 5% as tech device sales persist for now. We have long seen potential China restriction and the ramp of the region's local memory industry as MU's biggest threats, specifically from Yangtze Memory [YMTC] and ChangXin Memory. Still, MU has been able to successfully navigate "China storms" in the past [for instance, Huawei ban] and while the news will act as a revenue headwind over the next year, we think a 2H recovery remains intact. On a positive note, the outcome does ultimately help further derisk MU's exposure in China. / Angelo Zino, CFA

April 10, 2023

06:23 AM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 58.56****]:

Samsung Electronics says its cutting memory chip production by a "meaningful level" after reporting its lowest profit since the financial crisis, which we view as a notable positive for memory peers. We have been awaiting the move from the market share leader in NAND/DRAM chips for several months given announcements from other companies [e.g., MU cutting production by more than 40% in FY 23 and WFE spend by 50%] and should help end a massive supply glut in the coming months. We think the Samsung production cuts will help further normalize inventories and stabilize memory prices after declining more than 50% over the last 9 months, offering greater confidence toward a recovery in the second half. Although the pace of a recovery remains in question given macro trends, recent inventory write-downs by MU limit any further downside and should drive upside to depressed consensus views in the second half of the calendar year. Separately, China's cybersecurity review of MU, announced last week, does pose a new risk. /

Angelo Zino, CFA

March 29, 2023

12:29 AM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 59.28****]:

We keep our 12-month target price at \$68 on a P/E of 13.3x our FY 25 EPS view of \$5.11, below peers but above historical averages to reflect near trough fundamentals. We widen our FY 23 [Aug.] per share loss estimate to \$3.37 from a \$2.76 loss, largely on write-downs, and keep FY 24 at \$1.26 EPS. Sales fell 10% from Feb-Q [-53% Y/Y], near expectations, with NAND down 20% and DRAM falling a more modest 4%. Gross margin was -31.4%, hurt by \$1.4B of inventory write-downs [another \$500M seen in the May-Q] and steep underutilization, but is seen marking the cyclical trough. We do think efforts to sharply curtail production/capacity investments will help stabilize fundamentals, as we see orders sequentially steadily improving through the rest of the year. We believe DIOs peaked in the Feb-Q [153 days or 235 days ex. write-downs], which is historically a positive indicator. Looking ahead, we see MU benefiting from memory content growth in new CPU ramps, PC/smartphone declines leveling off, and China demand returning. / Angelo Zino, CFA

March 27, 2023

10:26 AM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 60.72****]:

We raise our 12-month target to \$68 from \$56 on a P/E of 13.3x our FY 25 EPS view of \$5.11, below peers but above historical averages to reflect near trough fundamentals. Ahead of Feb-Q results on March 28, we reduce our FY 23 [Aug.] per share estimate to a \$2.76 loss from \$1.96 loss and FY 24 to \$1.26 EPS from \$2.38. We expect a Feb-Q EPS loss of \$1.00 vs. \$2.16 EPS a year ago, as revenue is seen falling more than 50% Y/Y. Although customer inventories are improving, they are still elevated [areas like mobile are better than cloud] and pricing remains challenging. We do see risk of material inventory write-downs [lowers book value] and think revenue is likely to be pressured into the May-Q [revenue flat to down sequentially versus the Street up]. Still, we are encouraged by supply actions [-40% in FY 23, WFE spend -50%] and headcount cuts [-15% from -10% on the Nov-Q call]. Also, we see improving volumes in 2H, with prices firming as inventories improve, new CPUs ramp, and PC/smartphone declines level off. / Angelo Zino, CFA

December 22, 2022

07:08 AM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 49.70****]:

We cut our 12-month target to \$56 from \$63 on a P/E of 11x our FY 25 EPS view of \$5.11, near its 5-year historical forward average of 11.7x. We reduce our FY 23 [Aug.] per share estimate to a \$1.96 loss from \$0.46 EPS and FY 24 to \$2.38 EPS from \$4.61 EPS. MU posts Nov-Q loss per share of \$0.04 vs. \$2.16 EPS, wider than the \$0.02 consensus loss. Sales fell 39% sequentially, worse than expected, on a 41% plunge in DRAM and 35% drop in NAND [ASPs fell in the low 20% range]. Gross margin of 22.9% is seen compressing further to 8.5% in the Feb-Q, hurt by lower pricing/utilization. We have concerns about elevated inventories, with DIO at 214 days vs. 139 in Aug-Q [seen peaking in the Feb-Q]. However, we are encouraged by capex [down +50% in FY 23 and falling again in FY 24] and opex reductions [10% headcount cut]. We expect demand to gradually increase through CY 23 as customer inventories improve, new CPUs ramp, and China reopens. A supply/demand balance could be prolonged if Samsung doesn't notably cut capex. / Angelo Zino, CFA

November 16, 2022

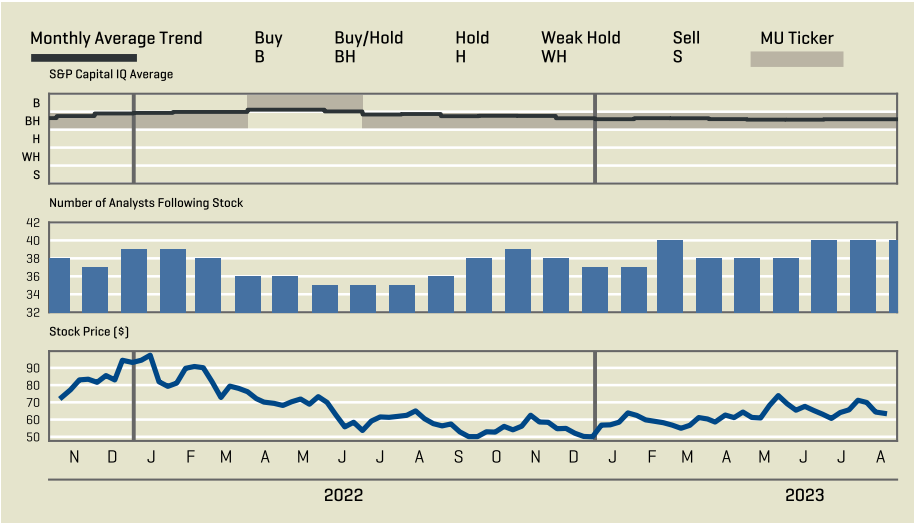
02:05 PM ET... CFRA Maintains Buy Opinion on Shares of Micron Technology, Inc. [MU 58.49****]:

MU says that it is reducing DRAM and NAND wafer starts by 20% vs. Aug-Q and working toward additional capex cuts. This follows comments in late September where MU said it was cutting capex by 30% in FY 23 and WFE spend by 50%. In CY 23, MU now expects its Y/Y bit supply growth to be negative for DRAM and less than 10% for NAND, significantly below prior expectations. We believe end-market dynamics [specifically on the consumer side] are likely to remain weaker for longer, creating the need for MU to more aggressively work down inventory. Given this, we see consensus estimates being revised down, especially for the next two to three quarters. That said, we think initiatives are necessary and will allow it to stabilize pricing and better position MU to see growth return by 2H CY 23, assuming macro conditions do not completely fall apart. We still expect fundamentals to trough in

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Micron Technology, Inc.

Analysts Recommendations



Wall Street Consensus Opinion

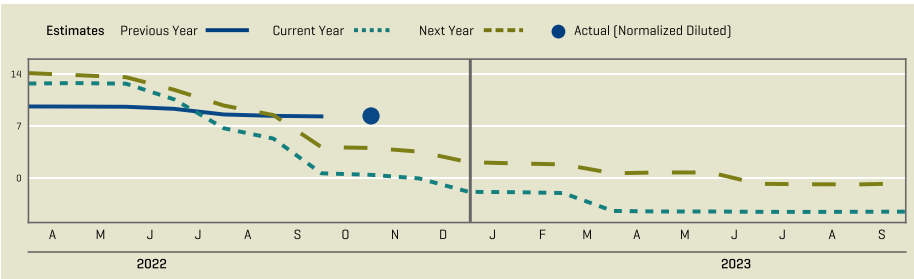
Buy/Hold

Wall Street Consensus vs. Performance

For fiscal year 2023, analysts estimate that MU will earn USD -4.54. For fiscal year 2024, analysts estimate that MU's earnings per share will grow by -83.85% to USD -0.73.

	No. of Recommendations	% of Total	1 Mo.Prior	3 Mos.Prior
Buy	20	50	18	17
Buy/Hold	8	20	8	7
Hold	7	18	10	10
Weak hold	1	3	1	1
Sell	2	5	1	1
No Opinion	2	5	2	2
Total	40	100	40	38

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2024	-0.73	2.86	-3.43	34	-94.07
2023	-4.54	-4.15	-4.79	33	-15.19
2024 vs. 2023	▲ 84%	▲ 169%	▲ 28%	▲ 3%	▼ -519%
Q4'24	0.49	2.03	-0.51	25	139.74
Q4'23	-1.18	-1.10	-1.27	30	-58.34
Q4'24 vs. Q4'23	▲ 142%	▲ 285%	▲ 60%	▼ -17%	▲ 340%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.

Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

Micron Technology, Inc.

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index [MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index]], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsize the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share [EPS] estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

CAGR - Compound Annual Growth Rate
 CAPEX - Capital Expenditures
 CY - Calendar Year
 DCF - Discounted Cash Flow
 DDM - Dividend Discount Model
 EBIT - Earnings Before Interest and Taxes
 EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
 EPS - Earnings Per Share
 EV - Enterprise Value
 FCF - Free Cash Flow
 FFO - Funds From Operations
 FY - Fiscal Year
 P/E - Price/Earnings
 P/NAV - Price to Net Asset Value
 PEG Ratio - P/E-to-Growth Ratio
 PV - Present Value
 R&D - Research & Development
 ROCE - Return on Capital Employed
 ROE Return on Equity
 ROI - Return on Investment
 ROIC - Return on Invested Capital
 ROA - Return on Assets
 SG&A - Selling, General & Administrative Expenses
 SOTP - Sum-of-The-Parts
 WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes [paid in the country of origin].

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



Micron Technology, Inc.

Disclosures

Stocks are ranked in accordance with the following ranking methodologies:

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Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five (six) model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section of the report for detailed methodology and the definition of Quantitative rankings.

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STARS Stock Reports:

Global STARS Distribution as of June 30, 2023

Ranking	North America	Europe	Asia	Global
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Hold	52.4%	48.0%	44.0%	49.8%
Sell	9.2%	15.3%	9.3%	10.4%
Total	100.0%	100.0%	100.0%	100.0%

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Micron Technology, Inc.

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